

Quotient rule

In calculus, the **quotient rule** is a method of finding the derivative of a function that is the ratio of two differentiable functions.^{[1][2][3]} Let $h(x) = \frac{f(x)}{g(x)}$, where both f and g are differentiable and $g(x) \neq 0$.

The quotient rule states that the derivative of $h(x)$ is

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

It is provable in many ways by using other derivative rules.

Examples

Example 1: Basic example

Given $h(x) = \frac{e^x}{x^2}$, let $f(x) = e^x$, $g(x) = x^2$, then using the quotient rule:

$$\begin{aligned} \frac{d}{dx} \left(\frac{e^x}{x^2} \right) &= \frac{\left(\frac{d}{dx} e^x \right) (x^2) - (e^x) \left(\frac{d}{dx} x^2 \right)}{(x^2)^2} \\ &= \frac{(e^x)(x^2) - (e^x)(2x)}{x^4} \\ &= \frac{x^2 e^x - 2x e^x}{x^4} \\ &= \frac{x e^x - 2e^x}{x^3} \\ &= \frac{e^x(x - 2)}{x^3}. \end{aligned}$$

Example 2: Derivative of tangent function

The quotient rule can be used to find the derivative of $\tan x = \frac{\sin x}{\cos x}$ as follows:

$$\begin{aligned}
\frac{d}{dx} \tan x &= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) \\
&= \frac{\left(\frac{d}{dx} \sin x \right) (\cos x) - (\sin x) \left(\frac{d}{dx} \cos x \right)}{\cos^2 x} \\
&= \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} \\
&= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\
&= \frac{1}{\cos^2 x} = \sec^2 x.
\end{aligned}$$

Reciprocal rule

The reciprocal rule is a special case of the quotient rule in which the numerator $f(x) = 1$. Applying the quotient rule gives

$$h'(x) = \frac{d}{dx} \left[\frac{1}{g(x)} \right] = \frac{0 \cdot g(x) - 1 \cdot g'(x)}{g(x)^2} = \frac{-g'(x)}{g(x)^2}.$$

Note that utilizing the [chain rule](#) yields the same result.

Proofs

Proof from derivative definition and limit properties

Let $h(x) = \frac{f(x)}{g(x)}$. Applying the definition of the derivative and properties of limits gives the following proof, with the term $f(x)g(x)$ added and subtracted to allow splitting and factoring in subsequent steps without affecting the value:

$$\begin{aligned}
h'(x) &= \lim_{k \rightarrow 0} \frac{h(x+k) - h(x)}{k} \\
&= \lim_{k \rightarrow 0} \frac{\frac{f(x+k)}{g(x+k)} - \frac{f(x)}{g(x)}}{k} \\
&= \lim_{k \rightarrow 0} \frac{f(x+k)g(x) - f(x)g(x+k)}{k \cdot g(x)g(x+k)} \\
&= \lim_{k \rightarrow 0} \frac{f(x+k)g(x) - f(x)g(x+k)}{k} \cdot \lim_{k \rightarrow 0} \frac{1}{g(x)g(x+k)} \\
&= \lim_{k \rightarrow 0} \left[\frac{f(x+k)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+k)}{k} \right] \cdot \frac{1}{g(x)^2} \\
&= \left[\lim_{k \rightarrow 0} \frac{f(x+k)g(x) - f(x)g(x)}{k} - \lim_{k \rightarrow 0} \frac{f(x)g(x+k) - f(x)g(x)}{k} \right] \cdot \frac{1}{g(x)^2} \\
&= \left[\lim_{k \rightarrow 0} \frac{f(x+k) - f(x)}{k} \cdot g(x) - f(x) \cdot \lim_{k \rightarrow 0} \frac{g(x+k) - g(x)}{k} \right] \cdot \frac{1}{g(x)^2} \\
&= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.
\end{aligned}$$

The limit evaluation $\lim_{k \rightarrow 0} \frac{1}{g(x+k)g(x)} = \frac{1}{g(x)^2}$ is justified by the differentiability of $g(x)$, implying continuity, which can be expressed as $\lim_{k \rightarrow 0} g(x+k) = g(x)$.

Proof using implicit differentiation

Let $h(x) = \frac{f(x)}{g(x)}$, so that $f(x) = g(x)h(x)$.

The product rule then gives $f'(x) = g'(x)h(x) + g(x)h'(x)$.

Solving for $h'(x)$ and substituting back for $h(x)$ gives:

$$\begin{aligned}
h'(x) &= \frac{f'(x) - g'(x)h(x)}{g(x)} \\
&= \frac{f'(x) - g'(x) \cdot \frac{f(x)}{g(x)}}{g(x)} \\
&= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.
\end{aligned}$$

Proof using the reciprocal rule or chain rule

Let $h(x) = \frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)}$.

Then the product rule gives $h'(x) = f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \frac{d}{dx} \left[\frac{1}{g(x)} \right]$.

To evaluate the derivative in the second term, apply the reciprocal rule, or the power rule along with the chain rule:

$$\frac{d}{dx} \left[\frac{1}{g(x)} \right] = -\frac{1}{g(x)^2} \cdot g'(x) = \frac{-g'(x)}{g(x)^2}.$$

Substituting the result into the expression gives

$$\begin{aligned} h'(x) &= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left[\frac{-g'(x)}{g(x)^2} \right] \\ &= \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{g(x)^2} \\ &= \frac{g(x)}{g(x)} \cdot \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{g(x)^2} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}. \end{aligned}$$

Proof by logarithmic differentiation

Let $h(x) = \frac{f(x)}{g(x)}$. Taking the absolute value and natural logarithm of both sides of the equation gives

$$\ln |h(x)| = \ln \left| \frac{f(x)}{g(x)} \right|$$

Applying properties of the absolute value and logarithms,

$$\ln |h(x)| = \ln |f(x)| - \ln |g(x)|$$

Taking the logarithmic derivative of both sides,

$$\frac{h'(x)}{h(x)} = \frac{f'(x)}{f(x)} - \frac{g'(x)}{g(x)}$$

Solving for $h'(x)$ and substituting back $\frac{f(x)}{g(x)}$ for $h(x)$ gives:

$$\begin{aligned}h'(x) &= h(x) \left[\frac{f'(x)}{f(x)} - \frac{g'(x)}{g(x)} \right] \\&= \frac{f(x)}{g(x)} \left[\frac{f'(x)}{f(x)} - \frac{g'(x)}{g(x)} \right] \\&= \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{g(x)^2} \\&= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.\end{aligned}$$

Note: Taking the absolute value of the functions is necessary for the logarithmic differentiation of functions that may have negative values, as logarithms are only real-valued for positive arguments. This works because $\frac{d}{dx}(\ln |u|) = \frac{u'}{u}$, which justifies taking the absolute value of the functions for logarithmic differentiation.

Higher order derivatives

Implicit differentiation can be used to compute the n th derivative of a quotient (partially in terms of its first $n - 1$ derivatives). For example, differentiating $f = gh$ twice (resulting in $f'' = g''h + 2g'h' + gh''$) and then solving for h'' yields

$$h'' = \left(\frac{f}{g} \right)'' = \frac{f'' - g''h - 2g'h'}{g}.$$

See also

- Chain rule – For derivatives of composed functions
- Differentiation of integrals – Problem in mathematics
- Differentiation rules – Rules for computing derivatives of functions
- General Leibniz rule – Generalization of the product rule in calculus
- Inverse functions and differentiation – Calculus identity
- Linearity of differentiation – Calculus property
- Product rule – Formula for the derivative of a product
- Reciprocal rule – differentiation rule
- Table of derivatives – Rules for computing derivatives of functions

- Vector calculus identities – Mathematical identities

References

1. Stewart, James (2008). *Calculus: Early Transcendentals* (https://archive.org/details/calculus_earlytra00stew_1) (6th ed.). Brooks/Cole. ISBN 978-0-495-01166-8.
 2. Larson, Ron; Edwards, Bruce H. (2009). *Calculus* (9th ed.). Brooks/Cole. ISBN 978-0-547-16702-2.
 3. Thomas, George B.; Weir, Maurice D.; Hass, Joel (2010). *Thomas' Calculus: Early Transcendentals* (12th ed.). Addison-Wesley. ISBN 978-0-321-58876-0.
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